



BAT MACHINE

French engineer Clément Ader took his inspiration for the Avion III flying machine from a bat. Like a bat's, the wings could be folded away for easy storage. However, this machine, built for the French Ministry of War, failed to become airborne when twice tested in 1897.

BIRDMAN PIONEER

German engineer Otto Lilienthal made over 2,000 flights in the 1890s using hang gliders of his own design. Although he was the author of an influential work on bird flight, Lilienthal believed that "a proper insight into the practice of flying" could only be achieved "by actual flying experiments."

foreshadowing of a helicopter.

If no one could see how to make a machine that would fly, they could possibly see why you would want to – especially in militaristic Europe, which was divided into states that were more or less permanently at war with one another. In 1670, proposing yet another impractical design – this time for an airship lifted by spheres from which the air had been pumped to create a vacuum – Italian Jesuit Father Francesco de Lana pointed out that such a vehicle could be used to land troops to capture a city in a surprise attack, or to destroy houses and fortresses by dropping “fireballs and bombs.”

Lighter-than-air flight

Although de Lana's vacuum-lifted airship was a nonstarter, it did point the way to the first successful human flight. De Lana's goal was “to make a machine lighter than the air itself.” This could not be done with vacuum spheres, but it could with a balloon filled with hot air or a light gas such as hydrogen. As usual in the history of invention, the solution to a problem became apparent to several inventors at once. When Joseph and Etienne Montgolfier, paper manufacturers from the French town of Annonay, brought a hot-air balloon to Paris in 1783, they faced competition from gentleman-scientist Jacques Charles, who was ready to demonstrate a hydrogen-filled balloon.



FIRST BALLOON FLIERS

In June 1783, the Montgolfier brothers conducted the first public display of a hot-air balloon, and the following November François Pilâtre de Rozier and the Marquis d'Arlandes made the first manned ascent (above). Early French aeronauts achieved some spectacular flights. In February 1784 Jean-Pierre Blanchard soared to over 12,500ft (3,800m) in a hydrogen balloon.